

Lab 1: ELF

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Task

- Learn how LDD utility works.
- Implement the app called **bldd** (backward ldd) – that shows all EXECUTABLE files that use specified shared library files. See example.
- App must be configurable (for ex. Possibility to set up scan directory)
- App must generate report as the output (txt, pdf, etc.) – can be implemented using any languages and libraries.
- Report must be sorted by number of executable usages (high -> low). See example.
- App must accept libraries with at least following arches: x86, x86_64, armv7, aarch64. (must be architecture dependent)
- App must have **help** with usage examples.

Solution

Here you can see the workflow of my program:

- Asciiinema URL: <https://asciiinema.org/a/ZZNUagjfP3ARxUY0>
- GitHub URL: <https://github.com/ilyalinhnguyen/adv-linux>

I implemented the task app in C (POSIX + direct ELF parsing). All tests I run in docker images with different arch.

The source code you can see at **bldd.c**.

Source code overview (**lab1/bldd.c**)

The program works like a reverse version of **ldd**:

- **Recursive scan**: it walks the selected directory tree using **nftw()** and checks each regular file.
- **ELF detection**: for each file it reads the first bytes and verifies ELF magic **0x7F 'E' 'L' 'F'**.
- **Architecture detection**: it reads **e_machine** and ELF class (32/64-bit) to classify binaries as:
 - **x86** (EM_386 + ELF32)
 - **x86_64** (EM_X86_64 + ELF64)
 - **armv7** (EM_ARM + ELF32 + EABI >= 5)
 - **aarch64** (EM_AARCH64 + ELF64)
- **Dependency extraction (DT_NEEDED)**:
 - it finds program headers (**PT_LOAD**, **PT_DYNAMIC**, **PT_INTERP**)

- reads the dynamic section and collects all `DT_NEEDED` entries
- finds the string table (`DT_STRTAB`) and resolves the `DT_NEEDED` offsets to library names
- **Filtering:** only keeps executables that depend on the libraries specified in CLI arguments.
- **Report building:** stores results grouped by architecture and library; for each library it stores unique executable paths.
- **Sorting:** libraries are sorted by number of executables (high → low) before printing.
- **Output formats:** supports `txt` (human readable) and `json` (machine readable) via `--format`.

Build with make

```
make
./bldd --help
```

Usage examples

Generate TXT report:

```
./bldd libc.so.6 libm.so.6 --scan-dir /usr/bin --arch x86_64 --output
report_usrbin.txt
```

Generate JSON report:

```
./bldd libc.so.6 --scan-dir /usr/sbin --arch x86_64 --format json --output
report_usrsbin.json
```

Docker demo (different architectures)

Build images (from repository root):

```
docker build -t bldd-amd64 -f ./lab1/Dockerfile.amd64 ./lab1
docker build -t bldd-arm64 -f ./lab1/Dockerfile.arm64 ./lab1
docker build -t bldd-aarch64 -f ./lab1/Dockerfile.aarch64 ./lab1
docker build -t bldd-armv7 -f ./lab1/Dockerfile.armv7 ./lab1
```

Run containers:

```
docker run --rm -it bldd-amd64
docker run --rm -it bldd-arm64
docker run --rm -it bldd-aarch64
docker run --rm -it bldd-armv7
```

Commands inside container (show full functionality):

```
uname -m
./bldd --help

# amd64 container:
./bldd libc.so.6 libm.so.6 --scan-dir /usr/bin --arch x86_64 --output
report_usrbin.txt

# arm64 / aarch64 container:
./bldd libc.so.6 libm.so.6 --scan-dir /usr/bin --arch aarch64 --output
report_usrbin.txt

# armv7 container:
./bldd libc.so.6 libm.so.6 --scan-dir /usr/bin --arch armv7 --output
report_usrbin.txt

# show JSON report:
./bldd libc.so.6 --scan-dir /usr/sbin --format json --output report.json
```

Screenshots

I will show only part of it, beacuse the output is very big, but the main functionality is working:

aarch64

```
[root@740d16c417f6:/app# uname -m
aarch64
[root@740d16c417f6:/app# ./bldd libc.so.6 libm.so.6 --scan-dir /usr/bin --arch aarch64 --output report_usrbin.txt
Report written to: report_usrbin.txt
[root@740d16c417f6:/app# cat report_usrbin.txt
Report on dynamic usage of selected shared libraries by ELF executables on /usr/bin
Target libraries: libc.so.6, libm.so.6
Generated at (UTC): 2026-05-06T08:58:09Z
```

```
----- aarch64 -----
libc.so.6 (263 execs)
-> /usr/bin/[
-> /usr/bin/aarch64-linux-gnu-addr2line
-> /usr/bin/aarch64-linux-gnu-ar
-> /usr/bin/aarch64-linux-gnu-as
-> /usr/bin/aarch64-linux-gnu-c++filt
-> /usr/bin/aarch64-linux-gnu-cpp-12
-> /usr/bin/aarch64-linux-gnu-dwp
-> /usr/bin/aarch64-linux-gnu-elfedit
-> /usr/bin/aarch64-linux-gnu-g++-12
-> /usr/bin/aarch64-linux-gnu-gcc-12
-> /usr/bin/aarch64-linux-gnu-gcc-ar-12
-> /usr/bin/aarch64-linux-gnu-gcc-nm-12
-> /usr/bin/aarch64-linux-gnu-gcc-ranlib-12
-> /usr/bin/aarch64-linux-gnu-gcov-12
-> /usr/bin/aarch64-linux-gnu-gcov-dump-12
-> /usr/bin/aarch64-linux-gnu-gcov-tool-12
-> /usr/bin/aarch64-linux-gnu-gp-archive
-> /usr/bin/aarch64-linux-gnu-gp-collect-app
```

```
[root@740d16c417f6:/app# ./bldd libc.so.6 libm.so.6 --scan-dir /usr/bin --format json --arch aarch64 --output report_usrbin.json
Report written to: report_usrbin.json
[root@740d16c417f6:/app# cat report_usrbin.json
{
  "scan_dir": "/usr/bin",
  "targets": ["libc.so.6", "libm.so.6"],
  "generated_at_utc": "2026-05-06T08:58:25Z",
  "report": {
    "aarch64": {
      "libc.so.6": ["/usr/bin/[", "/usr/bin/aarch64-linux-gnu-addr2line", "/usr/bin/aarch64-linux-gnu-ar", "/usr/bin/aarch64-linux-gnu-as", "/usr/bin/aarch64-linux-gnu-c++filt", "/usr/bin/aarch64-linux-gnu-cpp-12", "/usr/bin/aarch64-linux-gnu-dwp", "/usr/bin/aarch64-linux-gnu-elfedit", "/usr/bin/aarch64-linux-gnu-g++-12", "/usr/bin/aarch64-linux-gnu-gcc-12", "/usr/bin/aarch64-linux-gnu-gcc-ar-12", "/usr/bin/aarch64-linux-gnu-gcc-nm-12", "/usr/bin/aarch64-linux-gnu-gcc-ranlib-12", "/usr/bin/aarch64-linux-gnu-gcov-12", "/usr/bin/aarch64-linux-gnu-gcov-dump-12", "/usr/bin/aarch64-linux-gnu-gcov-tool-12", "/usr/bin/aarch64-linux-gnu-gp-archive", "/usr/bin/aarch64-linux-gnu-gp-collect-app", "/usr/bin/aarch64-linux-gnu-gprof", "/usr/bin/aarch64-linux-gnu-gprofng", "/usr/bin/aarch64-linux-gnu-ld.bfd", "/usr/bin/aarch64-linux-gnu-ld.gold", "/usr/bin/aarch64-linux-gnu-lto-dump-12", "/usr/bin/aarch64-linux-gnu-nm", "/usr/bin/aarch64-linux-gnu-objcopy", "/usr/bin/aarch64-linux-gnu-objdump", "/usr/bin/aarch64-linux-gnu-ranlib", "/usr/bin/aarch64-linux-gnu-readelf", "/usr/bin/aarch64-linux-gnu-size", "/usr/bin/aarch64-linux-gnu-strings", "/usr/bin/aarch64-linux-gnu-strip", "/usr/bin/addpart", "/usr/bin/apt", "/usr/bin/apt-cache", "/usr/bin/apt-cdrom", "/usr/bin/apt-config", "/usr/bin/apt-get", "/usr/bin/apt-mark", "/usr/bin/arch", "/usr/bin/b2sum", "/usr/bin/base32", "/usr/bin/base64", "/usr/bin/basename", "/usr/bin/basenc", "/usr/bin/bash", "/usr/bin/bunzip2", "/usr/bin/bzcat", "/usr/bin/bzip2", "/usr/bin/bzip2recover", "/usr/bin/cat", "/usr/bin/chage", "/usr/bin/chattr", "/usr/bin/chcon", "/usr/bin/chfn", "/usr/bin/chgrp", "/usr/bin/chmod", "/usr/bin/choom", "/usr/bin/chown", "/usr/bin/chrt", "/usr/bin/chsh", "/usr/bin/cksum", "/usr/bin/clear", "/usr/bin/clear_console", "/usr/bin/cmp", "/usr/bin/comm", "/usr/bin/cp", "/usr/bin/csplit", "/usr/bin/cut", "/usr/bin/dash", "/usr/bin/date", "/usr/bin/dd", "/usr/bin/delpart", "/usr/bin/df", "/usr/bin/diff", "/usr/bin/diff3", "/usr/bin/dir", "/usr/bin/dircolors", "/usr/bin/dirname", "/usr/bin/dmesg", "
```

x86_64

```
[~/Innopolis/adv-linux > docker run --rm -it bldd-amd64 12:00:11 ]
WARNING: The requested image's platform (linux/amd64) does not match the detected host platform (linux/arm64/v8) and no
specific platform was requested
[root@7b6ccfb4a8ab:/app# ls
Dockerfile.aarch64 Dockerfile.amd64 Dockerfile.arm64 Dockerfile.armv7 Makefile bldd bldd.c
[root@7b6ccfb4a8ab:/app# uname -m
x86_64
[root@7b6ccfb4a8ab:/app# ./bldd libc.so.6 libm.so.6 --scan-dir /usr/bin --arch x86_64 --output report_usrbin.txt
Report written to: report_usrbin.txt
[root@7b6ccfb4a8ab:/app# cat report_usrbin.txt
Report on dynamic usage of selected shared libraries by ELF executables on /usr/bin
Target libraries: libc.so.6, libm.so.6
Generated at (UTC): 2026-05-06T09:12:53Z

----- x86_64 -----
libc.so.6 (263 execs)
-> /usr/bin/[
-> /usr/bin/addpart
-> /usr/bin/apt
-> /usr/bin/apt-cache
-> /usr/bin/apt-cdrom
-> /usr/bin/apt-config
-> /usr/bin/apt-get
-> /usr/bin/apt-mark
-> /usr/bin/arch
-> /usr/bin/b2sum
-> /usr/bin/base32
-> /usr/bin/base64
```

```
[root@7b6ccfb4a8ab:/app# ./bldd libc.so.6 libm.so.6 --scan-dir /usr/bin --format json --arch x86_64 --output report_usrbi
n.json
Report written to: report_usrbin.json
[root@7b6ccfb4a8ab:/app# cat report_usrbin.json
{
  "scan_dir": "/usr/bin",
  "targets": ["libc.so.6", "libm.so.6"],
  "generated_at_utc": "2026-05-06T09:13:50Z",
  "report": {
    "x86_64": {
      "libc.so.6": ["/usr/bin/[", "/usr/bin/addpart", "/usr/bin/apt", "/usr/bin/apt-cache", "/usr/bin/apt-cdrom", "/usr/
bin/apt-config", "/usr/bin/apt-get", "/usr/bin/apt-mark", "/usr/bin/arch", "/usr/bin/b2sum", "/usr/bin/base32", "/usr/bi
n/base64", "/usr/bin/basename", "/usr/bin/basenc", "/usr/bin/bash", "/usr/bin/bunzip2", "/usr/bin/bzcat", "/usr/bin/bzip
2", "/usr/bin/bzip2recover", "/usr/bin/cat", "/usr/bin/chage", "/usr/bin/chattr", "/usr/bin/chcon", "/usr/bin/chfn", "/u
sr/bin/chgrp", "/usr/bin/chmod", "/usr/bin/chown", "/usr/bin/chrt", "/usr/bin/chsh", "/usr/bin/cksum",
"/usr/bin/clear", "/usr/bin/clear_console", "/usr/bin/cmp", "/usr/bin/comm", "/usr/bin/cp", "/usr/bin/csplit", "/usr/bi
n/cut", "/usr/bin/dash", "/usr/bin/date", "/usr/bin/dd", "/usr/bin/delpart", "/usr/bin/df", "/usr/bin/diff", "/usr/bin/d
iff3", "/usr/bin/dir", "/usr/bin/dircolors", "/usr/bin/dirname", "/usr/bin/dmesg", "/usr/bin/dpkg", "/usr/bin/dpkg-deb",
"/usr/bin/dpkg-divert", "/usr/bin/dpkg-query", "/usr/bin/dpkg-split", "/usr/bin/dpkg-statoverride", "/usr/bin/dpkg-trig
ger", "/usr/bin/du", "/usr/bin/echo", "/usr/bin/env", "/usr/bin/expand", "/usr/bin/expiry", "/usr/bin/expr", "/usr/bin/f
actor", "/usr/bin/faillog", "/usr/bin/fallocate", "/usr/bin/false", "/usr/bin/fincore", "/usr/bin/find", "/usr/bin/findm
nt", "/usr/bin/flock", "/usr/bin/fmt", "/usr/bin/fold", "/usr/bin/gencat", "/usr/bin/getconf", "/usr/bin/getent", "/usr/
bin/getopt", "/usr/bin/gpasswd", "/usr/bin/gpgv", "/usr/bin/grep", "/usr/bin/groups", "/usr/bin/gzip", "/usr/bin/hardlin
k", "/usr/bin/head", "/usr/bin/hostid", "/usr/bin/hostname", "/usr/bin/iconv", "/usr/bin/id", "/usr/bin/infocmp", "/usr/
bin/install", "/usr/bin/ionice", "/usr/bin/ipcmk", "/usr/bin/ipcrm", "/usr/bin/ipcs", "/usr/bin/ischroot", "/usr/bin/joi
n", "/usr/bin/last", "/usr/bin/lastlog", "/usr/bin/link", "/usr/bin/ln", "/usr/bin/locale", "/usr/bin/localedef", "/usr/
bin/logger", "/usr/bin/login", "/usr/bin/logname", "/usr/bin/ls", "/usr/bin/lsattr", "/usr/bin/lsblk", "/usr/bin/lscpu",
"/usr/bin/lsfd", "/usr/bin/lsipc", "/usr/bin/lsmem", "/usr/bin/lslocks", "/usr/bin/lslogins", "/usr/bin/lsnmap", "/usr/b
in/lsns", "/usr/bin/lzmainfo", "/usr/bin/make", "/usr/bin/mawk", "/usr/bin/mcookie", "/usr/bin/md5sum", "/usr/bin/mesg",
"/usr/bin/mkdir", "/usr/bin/mkfifo", "/usr/bin/mknod", "/usr/bin/mktemp", "/usr/bin/more", "/usr/bin/mount", "/usr/bin/
```

armv7

```

~/Innopolis/adv-linux > docker run --rm -it bldd-armv7 11:59:12 ]
root@3067b1ecf5a1:/app# ls ]
Dockerfile.aarch64 Dockerfile.amd64 Dockerfile.arm64 Dockerfile.armv7 Makefile bldd bldd.c ]
root@3067b1ecf5a1:/app# uname -m ]
armv7l ]
root@3067b1ecf5a1:/app# ./bldd --help ]
Usage: ./bldd [options] <library> [library...]

Backward ldd: find ELF executables that depend on selected shared libraries.

Options:
  -d, --scan-dir DIR      Directory to recursively scan (default: .)
  -a, --arch ARCH         x86 | x86_64 | armv7 | aarch64 (repeatable)
      --include-hidden    Include hidden files/directories
  -f, --format FORMAT     txt | json (default: txt)
  -o, --output FILE       Output report file (default: bldd_report.txt)
  -h, --help              Show this help

Examples:
  ./bldd libc.so.6 --scan-dir /usr/bin --output report.txt
  ./bldd libc.so.6 libm.so.6 --scan-dir /home --arch x86_64 --arch aarch64
  ./bldd /lib/x86_64-linux-gnu/libc.so.6 --scan-dir / --format json --output report.json
root@3067b1ecf5a1:/app# ./bldd libc.so.6 libm.so.6 --scan-dir /usr/bin --arch armv7 --output report_usrbin.txt ]
Report written to: report_usrbin.txt ]
root@3067b1ecf5a1:/app# cat report_usrbin.txt ]
Report on dynamic usage of selected shared libraries by ELF executables on /usr/bin
Target libraries: libc.so.6, libm.so.6
Generated at (UTC): 2026-05-06T08:59:36Z

----- armv7 -----

```

```

----- armv7 -----
libc.so.6 (258 execs)
-> /usr/bin/[
-> /usr/bin/addpart
-> /usr/bin/apt
-> /usr/bin/apt-cache
-> /usr/bin/apt-cdrom
-> /usr/bin/apt-config
-> /usr/bin/apt-get
-> /usr/bin/apt-mark
-> /usr/bin/arch
-> /usr/bin/arm-linux-gnueabi-hf-addr2line
-> /usr/bin/arm-linux-gnueabi-hf-ar
-> /usr/bin/arm-linux-gnueabi-hf-as
-> /usr/bin/arm-linux-gnueabi-hf-c++filt
-> /usr/bin/arm-linux-gnueabi-hf-cpp-12
-> /usr/bin/arm-linux-gnueabi-hf-dwp
-> /usr/bin/arm-linux-gnueabi-hf-elfedit
-> /usr/bin/arm-linux-gnueabi-hf-g++-12
-> /usr/bin/arm-linux-gnueabi-hf-gcc-12
-> /usr/bin/arm-linux-gnueabi-hf-gcc-ar-12
-> /usr/bin/arm-linux-gnueabi-hf-gcc-nm-12
-> /usr/bin/arm-linux-gnueabi-hf-gcc-ranlib-12
-> /usr/bin/arm-linux-gnueabi-hf-gcov-12
-> /usr/bin/arm-linux-gnueabi-hf-gcov-dump-12
-> /usr/bin/arm-linux-gnueabi-hf-gcov-tool-12
-> /usr/bin/arm-linux-gnueabi-hf-gprof
-> /usr/bin/arm-linux-gnueabi-hf-ld.bfd
-> /usr/bin/arm-linux-gnueabi-hf-ld.gold

```

```
[root@3067b1ecf5a1:/app# ./bldd libc.so.6 libm.so.6 --scan-dir /usr/bin --format json --arch armv7 --output report_usrbin.json
Report written to: report_usrbin.json
[root@3067b1ecf5a1:/app# cat report_usrbin.json
{
  "scan_dir": "/usr/bin",
  "targets": ["libc.so.6", "libm.so.6"],
  "generated_at_utc": "2026-05-06T08:59:48Z",
  "report": {
    "armv7": {
      "libc.so.6": ["/usr/bin/[, "/usr/bin/addpart", "/usr/bin/apt", "/usr/bin/apt-cache", "/usr/bin/apt-cdrom", "/usr/bin/apt-config", "/usr/bin/apt-get", "/usr/bin/apt-mark", "/usr/bin/arch", "/usr/bin/arm-linux-gnueabi-hf-addr2line", "/usr/bin/arm-linux-gnueabi-hf-ar", "/usr/bin/arm-linux-gnueabi-hf-as", "/usr/bin/arm-linux-gnueabi-hf-c++filt", "/usr/bin/arm-linux-gnueabi-hf-cpp-12", "/usr/bin/arm-linux-gnueabi-hf-dwp", "/usr/bin/arm-linux-gnueabi-hf-elfedit", "/usr/bin/arm-linux-gnueabi-hf-g++-12", "/usr/bin/arm-linux-gnueabi-hf-gcc-12", "/usr/bin/arm-linux-gnueabi-hf-gcc-ar-12", "/usr/bin/arm-linux-gnueabi-hf-gcc-nm-12", "/usr/bin/arm-linux-gnueabi-hf-gcc-ranlib-12", "/usr/bin/arm-linux-gnueabi-hf-gcov-12", "/usr/bin/arm-linux-gnueabi-hf-gcov-dump-12", "/usr/bin/arm-linux-gnueabi-hf-gcov-tool-12", "/usr/bin/arm-linux-gnueabi-hf-gprof", "/usr/bin/arm-linux-gnueabi-hf-ld.bfd", "/usr/bin/arm-linux-gnueabi-hf-ld.gold", "/usr/bin/arm-linux-gnueabi-hf-lto-dump-12", "/usr/bin/arm-linux-gnueabi-hf-nm", "/usr/bin/arm-linux-gnueabi-hf-objcopy", "/usr/bin/arm-linux-gnueabi-hf-objdump", "/usr/bin/arm-linux-gnueabi-hf-ranlib", "/usr/bin/arm-linux-gnueabi-hf-readelf", "/usr/bin/arm-linux-gnueabi-hf-size", "/usr/bin/arm-linux-gnueabi-hf-strings", "/usr/bin/arm-linux-gnueabi-hf-strip", "/usr/bin/b2sum", "/usr/bin/base32", "/usr/bin/base64", "/usr/bin/basename", "/usr/bin/basenc", "/usr/bin/bash", "/usr/bin/bunzip2", "/usr/bin/bzcat", "/usr/bin/bzip2", "/usr/bin/bzip2recover", "/usr/bin/cat", "/usr/bin/chage", "/usr/bin/chattr", "/usr/bin/chcon", "/usr/bin/chfn", "/usr/bin/chgrp", "/usr/bin/chmod", "/usr/bin/chown", "/usr/bin/chrt", "/usr/bin/chsh", "/usr/bin/cksum", "/usr/bin/clear", "/usr/bin/clear_console", "/usr/bin/cmp", "/usr/bin/comm", "/usr/bin/cp", "/usr/bin/csplit", "/usr/bin/cut", "/usr/bin/dash", "/usr/bin/date", "/usr/bin/dd", "/usr/bin/delpart", "/usr/bin/df", "/usr/bin/diff", "/usr/bin/diff3", "/usr/bin/dir", "/usr/bin/dircolors", "/usr/bin/dirname", "/usr/bin/dmesg", "/usr/bin/dpkg", "/usr/bin/dpkg-deb", "/usr/bin/dpkg-divert", "/usr/bin/dpkg-query", "/usr/bin/dpkg-split", "/usr/bin/dpkg-statoverride", "/usr/bin/dpkg-trigger", "/usr/bin/du", "/usr/bin/echo", "/usr/bin/env", "/usr/bin/expand", "/usr/bin/expiry", "/usr/bin/expr", "/usr/bin/factor
```

How it works in practical you can see in Asciiinema earlier.